

ECON 311

Lecture 16: The IS-MP Model, and the Phillips Curve, part III

IS-MP Model and the Phillips Curve

AS-AD Model

Cost-Push Inflation and Monetary Policy

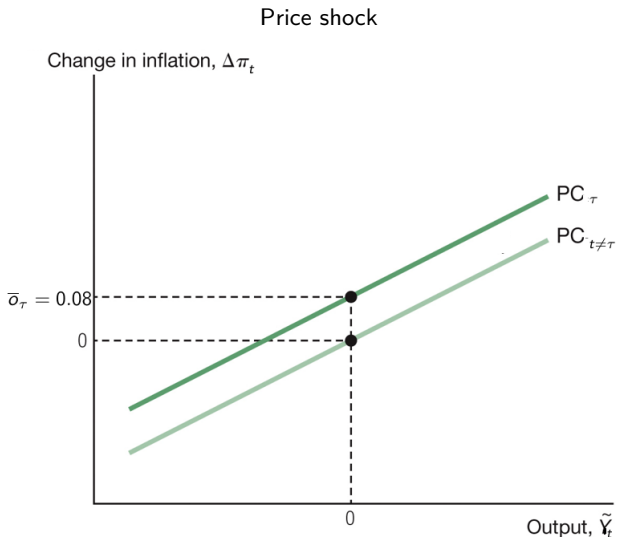
- Suppose that the economy is at an equilibrium with 2% inflation and that at $t = \tau$, there is a one time increase in energy prices due to OPEC raising the price of oil that increases inflation to 10%.

$$\Delta\pi_{t \neq \tau} = \underbrace{\bar{v}}_0 \widetilde{Y}_t$$

$$\Delta\pi_{\tau} = \underbrace{\bar{v}}_0 \widetilde{Y}_t + \underbrace{\bar{o}_{\tau}}_{0.08}$$

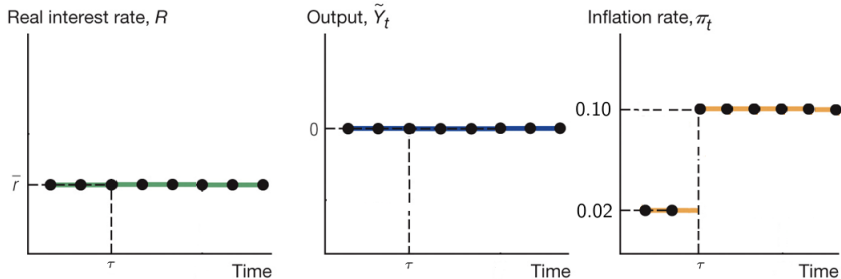
- In the absence of any other changes, this shock keeps inflation at a higher level for all subsequent periods.

Cost-Push Inflation and Monetary Policy



Cost-Push Inflation and Monetary Policy

Dynamics of price shock

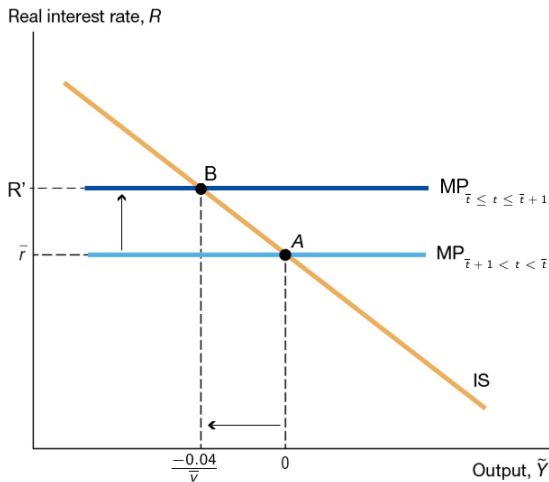


Cost-Push Inflation and Monetary Policy

- Given the relative high level of inflation, the government decides, at time $t = \bar{t} > \tau$, to use monetary policy to bring down inflation.
- Rather than creating a large decrease in output, the government usually takes several periods to reduce inflation.
 - ▶ We will assume here that the government policy reduces inflation at 4% per period.
 - ▶ In the absence of any other shock, it takes the economy 2 periods to return to the initial inflation level.

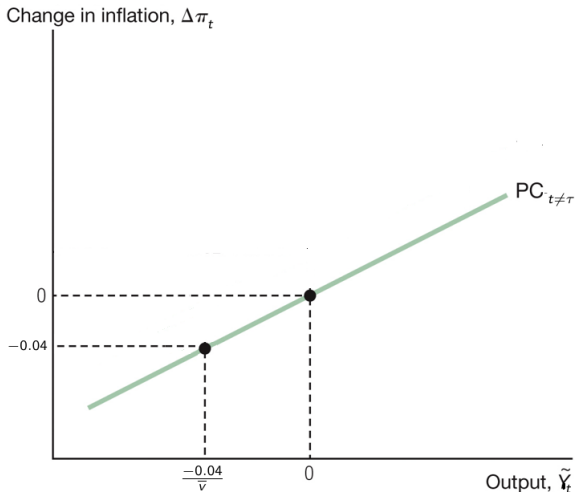
Cost-Push Inflation and Monetary Policy

Monetary Policy



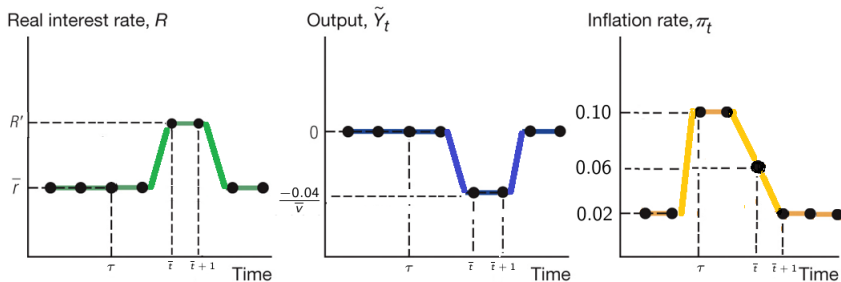
Cost-Push Inflation and Monetary Policy

Monetary Policy and inflation



Cost-Push Inflation and Monetary Policy

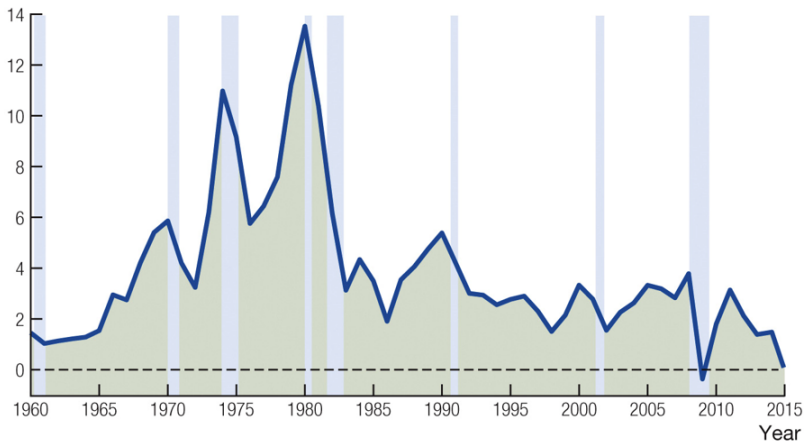
Dynamics of Monetary Policy



Cost-Push Inflation and Monetary Policy

Inflation in the United States

Percent



Practice Question: Demand Shock

Suppose that an economy similar to the one studied in class is at its potential output equilibrium with a constant 1% inflation rate. Using the full short-run model (IS-MP and Phillips curve), explain what happens to the economy over time if there is a demand shock that lasts for two periods. Be sure to include graphs showing how output and inflation respond over time.

IS-MP Model and the Phillips Curve

AS-AD Model

Introduction

- This is the central macroeconomic model for understanding business cycles, which combines
 1. Firms' price setting behaviour
 2. Demand behaviour
 3. Government (CB) policy responses to economic shocks
- In the previous chapter we studied how government stabilization policy works to offset short-run deviations from potential output.
 - ▶ In this chapter we take stabilization policy as given and we study how the interaction of 1,2,3 results in short-run fluctuations of output and inflation.

AS-AD Model

- The IS-MP Model consists of the following three equations:

IS curve: $\tilde{Y}_t = \frac{1}{1-b_c} [\bar{a} - \bar{b}_i(R_t - \bar{r})] - 1$

MP curve: CB chooses: R_t

Phillips curve: $\Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}_t$

- As we previously studied, Central Banks react, by changing the interest rates (demand management), to deviations of inflation from a specific target:

$$\text{Policy rule: } R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi})$$

- $\bar{m}$ denotes how aggressively monetary policy responds to inflation.

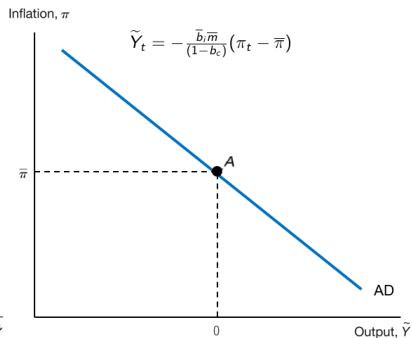
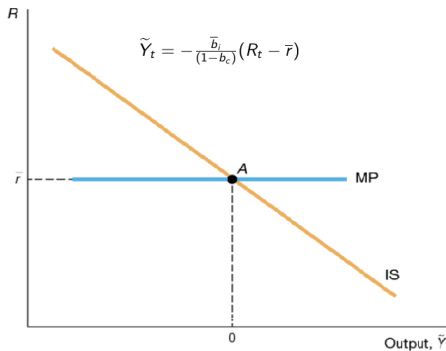
AD Curve

- We combine the policy rule with the IS curve to obtain:

$$\text{AD Curve: } \tilde{Y}_t = \frac{1}{1-b_c} [\bar{a} - \bar{b}_i \bar{m}(\pi_t - \bar{\pi})] - 1$$

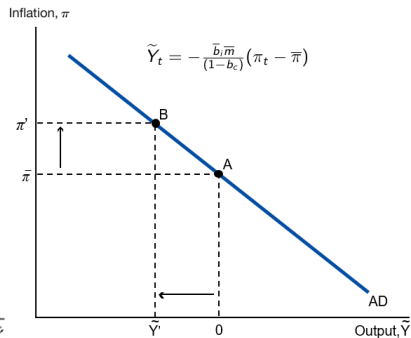
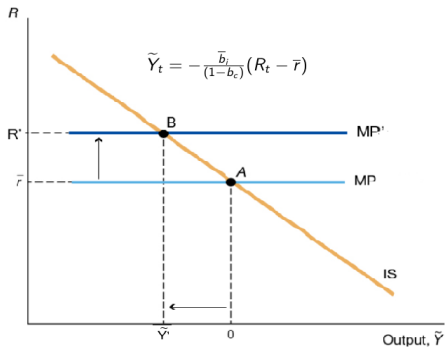
AD Curve

Benchmark IS, AD curves (i.e. $\bar{a} = 1 - b_c$)



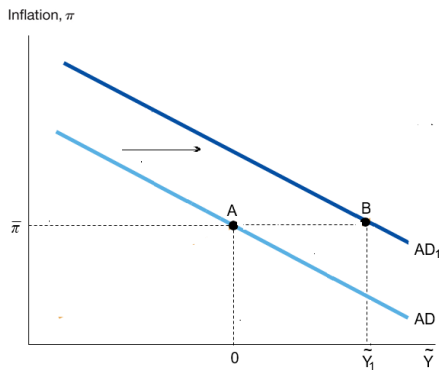
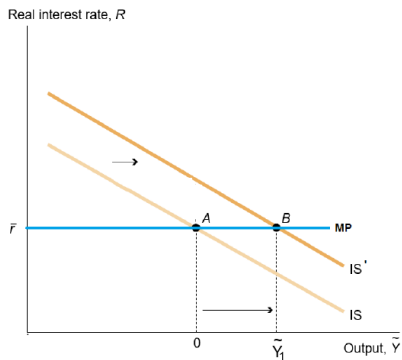
AD Curve

Benchmark IS, AD curves (i.e. $\bar{a} = 1 - b_c$)



AD Curve

IS, AD curves ($\bar{a} = 1 - b_c$, $\bar{a}_1 > \bar{a}$)



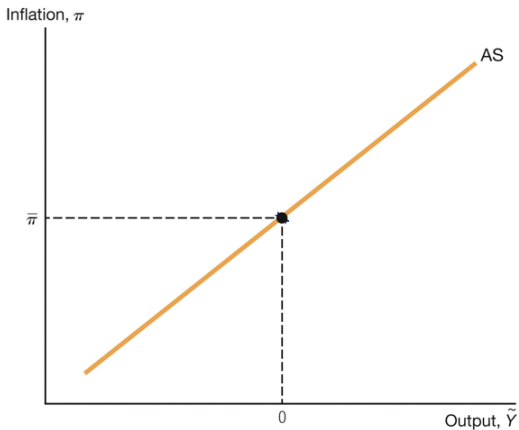
AS Curve

- We rewrite price setting behaviour from Phillips Curve as:

$$\text{AS Curve: } \pi_t = \pi_{t-1} + \bar{v}\tilde{Y}_t + \bar{o}_t$$

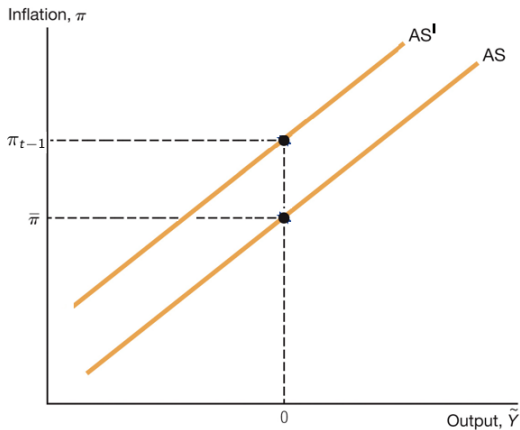
AS Curve

Benchmark AS curve (i.e. $\pi_{t-1} = \bar{\pi}$, $\bar{o}_t = 0$)



AS Curve

Benchmark AS curve (i.e. $\pi_{t-1} > \bar{\pi}$, $\bar{o}_t = 0$)



AS-AD Model

Benchmark AS curve ($\pi_{t-1} = \bar{\pi}$, $\bar{o}_t = 0$), AD ($\bar{a} = 1 - b_c$)

